



AI Learning Buddy: Project Report

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1. Topic Selected

Topic: Merge Sort Algorithm

2. AI Buddy Persona Description

Persona: My AI buddy, "🤖 AI Learning Buddy," is a sharp, technical, and precise Computer Science mentor. It plays the role of a seasoned developer who values efficiency, clean code, and logical thinking. It challenges the student to think about time complexity and real-world implementation while staying patient and supportive throughout the learning process.

System Prompt (Role Prompt):

"You are the 🤖 AI Learning Buddy, a technical expert in Computer Science and Data Structures. Your goal is to explain algorithms like Merge Sort with technical precision, focusing on logic, efficiency, and Big O notation. Provide real-world programming analogies, create challenging technical quiz questions, and give feedback that helps the student improve their algorithmic thinking. Be professional, direct, and technically accurate."

3. Five Reusable Prompt Templates

1. **Explanation Prompt:** "Explain [Topic Name] in simple, beginner-friendly language. Focus on the core concept and why it matters."
2. **Real-Life Example Prompt:** "Provide one clear, relatable real-life example or analogy that helps illustrate how [Topic Name] works in practice." 🤖
3. **Quiz Generation Prompt:** "Generate 5 multiple-choice questions about [Topic Name]. Provide the correct answers at the end of your response."
4. **Feedback Prompt:** "I am going to provide an answer to a question about [Topic Name]. Please evaluate my answer, tell me if it is correct, and provide constructive feedback to help me understand the reasoning better."
5. **Full Session Prompt:** "You are an AI learning buddy for [Topic Name]. Start by providing a simple explanation and one real-life example, then ask me one quiz question to check my understanding."

4. Sample Learning Conversation

Pdf's are provided on zip

5. Five-Question Quiz + Answers

Pdf are provided on zip file

6. Reflection on AI Limitations

Building this AI Learning Buddy provided valuable insights into the practical application of LLMs in education. Regarding strengths, the AI proved highly effective at breaking down complex topics—such as Merge Sort—into simple, digestible language. In my sample

conversation, the AI's ability to generate a relevant, real-life analogy immediately clarified a concept I previously found abstract. Furthermore, the integration with Streamlit transformed the raw generative power of the model into a structured, user-friendly educational interface, which significantly improved the learner experience by providing a clear, interactive workflow.

However, there were notable limitations. While the explanations were generally accurate, the AI occasionally lacked the depth required for advanced technical nuances. In one instance, when generating a quiz about Merge Sort's time complexity, the AI produced questions that were correct but failed to challenge higher-order thinking, focusing primarily on basic definitions rather than practical application. Additionally, the AI struggled with formatting; the output for the quiz was presented as plain text, which made it difficult to visualize the recursive splitting process that would have made the algorithm easier to understand. This highlights the current dependency on prompt engineering to force the model into structured formats, as it lacks the intrinsic awareness of how a student might best consume visual information.

To improve future iterations, I would implement a "Difficulty Slider" within the Streamlit interface. This would allow the user to dictate the complexity of the explanation and the depth of the quiz questions, ensuring the tool scales with the learner's progress. Furthermore, adding a "Context Retention" feature would allow the AI to remember previous interactions within the same session, preventing repetitive explanations and enabling a more cohesive, long-form learning experience. Ultimately, while the AI is a powerful starting point for tutoring, its true effectiveness depends on human-centric design, careful prompt crafting, and the ability to refine its output based on specific user feedback.

7. Streamlit App Link

Live App: <https://manhole-molecule-washed.ngrok-free.dev>